

Real-Time Fraud Detection System with LLM-Assisted Risk Analysis

Project Overview

This project develops a real-time fraud detection system designed for financial transaction monitoring. The system integrates machine learning models with a large language model (LLM) reasoning layer to identify potentially fraudulent behavior, generate interpretable explanations, and support decision-making for different stakeholders. Unlike traditional systems that only output a risk score, this system emphasizes both predictive accuracy and explainability, enabling analysts, product managers, and business users to understand not just what is predicted, but why.

System Architecture and Workflow

The system begins with a data ingestion layer that collects transaction-level data, including transaction amount, timestamp, geolocation, device information, and historical user behavior. Additional external signals such as IP risk scores and blacklist indicators are also incorporated. This data is then processed through a feature engineering pipeline, where behavioral features such as transaction frequency, deviation from historical patterns, velocity metrics, and temporal features are constructed to capture abnormal activity.

The processed features are passed into a model layer consisting of both tree-based models, such as Random Forest and XGBoost, and sequence-based models such as LSTM. These models generate a probability score indicating the likelihood of fraud. The outputs of these models, along with relevant contextual features, are then passed to an LLM-based reasoning layer. This layer translates numerical predictions into structured, human-readable explanations by linking observed anomalies to potential risk factors, such as unusual location changes or abnormal transaction volumes.

Finally, a decision layer uses configurable thresholds and business rules to determine whether a transaction should be approved, flagged for review, or blocked. The system is designed to operate under real-time constraints, ensuring that predictions and explanations are generated within milliseconds to support live transaction processing.

Technical Details

To address the extreme class imbalance inherent in fraud detection, where fraudulent transactions typically represent less than 1% of all data, the system applies techniques such as weighted loss functions and oversampling methods like SMOTE. Model performance is evaluated using metrics such as ROC-AUC, precision-recall curves, and false positive rates, with particular emphasis on balancing detection accuracy against customer experience. Latency optimization is achieved through batch inference, caching of frequently accessed features, and efficient model serving strategies.

Key Challenges

The system faces several critical challenges. First, the imbalance in the dataset makes it difficult to detect rare fraud cases without generating excessive false positives. Second, there is a fundamental trade-off between fraud detection sensitivity and user experience, as overly aggressive models may block legitimate transactions. Third, the system must meet strict real-time latency requirements, which limits the complexity of models that can be deployed. Finally, ensuring that model predictions are interpretable and trustworthy is essential for adoption by business stakeholders.

Limitations

Despite its strengths, the system has several limitations. It relies heavily on historical fraud patterns, which means it may struggle to detect entirely new types of fraud. Additionally, the explanations

generated by the LLM layer are approximations and may not always perfectly reflect the internal logic of the underlying models. Data drift over time can also degrade model performance, requiring periodic retraining and monitoring. Furthermore, external risk signals may introduce noise, leading to unstable predictions in certain cases.

Business Impact

From a business perspective, the system provides significant value by reducing fraud-related financial losses, improving customer trust, and increasing operational efficiency. By providing interpretable explanations alongside predictions, the system enables faster and more confident decision-making by analysts. It also supports scalable fraud prevention strategies by automating large portions of the review process while maintaining transparency.

Future Improvements

Future enhancements to the system include implementing online learning to adapt to evolving fraud patterns in real time, incorporating graph-based methods to capture relationships between accounts and transactions, and introducing human-in-the-loop feedback mechanisms to continuously refine model performance. Additionally, cost-sensitive optimization techniques can be applied to better balance fraud risk against customer experience, and further improvements to the LLM reasoning layer can enhance the alignment between explanations and model behavior.